

Staff Days — Week 1, Day 2 (Tue Jul 28)

Joe's Morning Session — *Brain Lifts, AI, and Learning Science*

0:18–2:11

Joe opened by flagging the format as deliberate irony: a domain expert lecturing 350 people is "the collection of all the worst practices combined," and he predicted the room would retain 5–10 ideas out of two and a half hours. He ran live checks for understanding to make the point — most of the room didn't know Project Follow Through, DOK3, or direct instruction, and only ~10% knew gradient descent.

Brain lifts

A **brain lift** is the data structure for human knowledge — a written description of what's actually in your head. It must be **typed by a human**. AI-generated brain lifts are the tell that nothing is in your brain, and a tool that was generating them for students got shut down.

Purpose: without a brain lift loaded on top of it, an LLM only gives you the average of the internet. "If you can solve your problem with a prompt, so can everybody else in the world."

Joe's stated hiring bar for the summer interns: build a great brain lift and the job is guaranteed. Expectation for non-guide staff is **an hour a day** maintaining your own knowledge. He acknowledged guides can't carve that out and said high school would develop a compressed way to teach it.

Cognitive onload vs. offload

Chatbots were built for **offload**, which is why "90% of chatbots used in schools are used as cheatbots." Learning requires **onload**. The teaching problem is helping kids know which is which.

Depth of Knowledge

		AI capability
DOK1	Facts	Very good
DOK2	Summaries	Very good
DOK3	Insights on a fact base	Decent on frontier models

		AI capability
DOK4	Creating new knowledge	Terrible — this is the human moat

You cannot reach DOK3 or 4 without DOK1 loaded in long-term memory. On the "kids don't need facts anymore" argument: an AI reasoning without a fact base is called hallucination; a student doing it is making things up. **Critical reasoning only happens over a fact base.**

Where humans beat LLMs

Knowing which question matters; holding two conflicting ideas at once (LLMs are consensus mechanisms that drive to the average); rebuilding a field from first principles; integrating across domains. Test for a real spiky point of view: if the AI instantly agrees, it isn't spiky. If it calls you unethical, you're onto something.

Truthseeker, not "critical thinker"

Critical thinking has become meaningless. The replacement is **truthseeker** — a daily discipline of trying to falsify your own beliefs, not an attitude. Middle school application: steelman the opposing side, or your idea gets shelved.

Philosopher builders

The high school output target. Philosophy because AI forces the existential questions back to the center; builder because AI can't run experiments or generate new real-world data. Cited example: an Alpha High student with a Nature paper due to publish in 4–6 weeks, showing a trained AI gives better social-emotional advice to teens than ChatGPT *and* than parents. Joint research slots signed with MIT and Harvard — up to 30 students each, ~60 total.

Learning science positions worth knowing

- **If a student isn't learning, it's the system's fault, not the student's.** Fewer than 10% of Alpha kids spend over 2 hours and learn under 2X; those average roughly 2.5 hours for 1.6X.
- **Mastery is the least understood and most important concept.** Median US 8th grader scores 233 on MAP; median 12th grader scores 234 — one point in four years.
- **Direct instruction vs. inquiry learning:** you need both, but most schools invert them. Load knowledge fast via direct instruction, then spend the freed time on inquiry-based life skills workshops. Direct instruction is ~5x faster for the average kid, and when kids are offered the trade they pick it.
- **Motivation over IQ.** The two predictors in a conveyor-belt system are IQ and Big Five conscientiousness. Mastery solves for IQ; motivation substitutes for conscientiousness.

- **DOP framing tip:** parents shrug at "51st percentile" but react strongly to "two grade levels behind" and even more to "can't answer 4th grade questions." Sports analogies work — master the basics before the alley-oop.
- **Apps:** Math Raiders replaces Alpha Math (faster *and* more fun; Trash Cat was cut for being fun but too slow). AlphaTalk ships in a TikTok-style and a plain full-screen version, both testing in the next two weeks.

Lunch and Learn follow-up (2:19–2:54)

Can AI build your brain lift? No — writing is thinking, and the document must be typed by you. AI is fine for finding papers, arguing against you, or asking "what won't you need in this brain lift in 12 months." Keep brain lifts **narrow**; multi-domain ones confuse the model. Teaching it to kids means starting from what they already love (the AI video workshop was the worked example) and letting DOK1 → DOK2 → DOK3 build naturally. How young to start is an open question the guides have to figure out.

Post-Lunch Tee-Up — *AlphaCore* → Beat the Bangers

3:04–3:22

Steph and Ben introduced **AlphaCore** as the Common Core for life skills — the system that teaches, defines, and evaluates them. The framing reason it matters now: TimeBack is commoditized and available outside Alpha, so **the afternoon is the product**.

The 3D Kid — Me / We / Key

The rationale for the five priority skills (grit & hard work, financial literacy & entrepreneurship, teamwork & leadership, relationship building & socialization, public speaking & storytelling), presented through what goes wrong when a dimension is missing:

- **Me + Key, no We** → brilliant but lonely. Wins the science fair, no one to celebrate with.
- **We + Key, no Me** → popular but unproductive. Charming, can't sit still for a hard thing.
- **Me + We, no Key** → perfect today, obsolete tomorrow. Straight A's, never shipped anything real.

Kids can't level up without mastery in every dimension. ("Key" is a placeholder — Steph asked the room for a better word.)

Bangers

A banger is a workshop kids love, that has buzz and agency, that's nationally normed so parents recognize it, grounded in a real life skill, engaged with the real world, and showable.

Every one needs an **alpha test** — a performance bar, not a rubric. Existing stacks run vertically (grit: 40-foot rock wall → 5-mile bike → Spartan race) and horizontally by level theme.

Build assignment

Breakouts by level group to **beat the existing bangers**, then a 4pm pitch-a-thon. The stated bar: make Joe visibly squeamish, the way Faith did proposing second graders run a sub-35-minute 5K.

What cleared the bar: LL beekeepers running a real apiary and selling the honey; kindergartners building an Opti sailboat their parents then have to sit in; a Super Sprint triathlon at the 13–15-year-old elite distance; L1 swimming a mile without stopping; middle schoolers building and flying a plane; middle schoolers dropped on a private island in Thailand for a MrBeast video. Joe's repeated note: the banger only lands if you can compress it into one line that unsettles a parent. He also flagged the island one as a natural ladder into World School.

Also announced: Alpha High's Session 1 boot camp puts transfer students through K–8 life skill tests, including receiving critical feedback from their parents played in front of the student body with a heart-rate-return-to-baseline standard.